

SHAHARIAR HOSSAIN MAHIR

Privacy-Preserving AI • Federated Learning • Foundation Models • Trustworthy LLMs

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Research Profile

I work at the intersection of **privacy-preserving machine learning**, **federated learning**, **foundation models**, and **trustworthy AI**. My research focuses on building scalable collaborative learning systems that remain robust under heterogeneous data, preserve privacy, and provide interpretable predictions for high-impact domains such as healthcare and climate analytics. Current work extends to latent-space fusion for federated foundation models, federated semi-supervised crop disease diagnosis (AgriSSFL), and security-oriented analysis of large language models.

Research Highlights

- **Cite:** 216 | **h-index:** 8 | **i10-index:** 6
- 8 published Q1 journal papers
- 3 first-author journal papers
- 3 first-author manuscripts under review/submission
- International research grants and awards
- Research across FL, XAI, FM, and LLM Security

Research Contributions

- Developed federated learning pipelines for decentralized and heterogeneous datasets, with emphasis on privacy, reproducibility, and non-IID optimization.
- Investigated foundation-model adaptation and latent representation fusion to improve scalability under distributed learning constraints.
- Developed **AgriSSFL**, a federated semi-supervised learning framework for crop disease diagnosis that jointly addresses class imbalance, non-IID data, and label scarcity.
- Studied jailbreak vulnerabilities and countermeasures for large language models, contributing to trustworthy AI evaluation methodologies.

EDUCATION

University of Dhaka

2020–2025

B.Sc. in Computer Science and Engineering

Thesis: *Privacy-Preserving Federated Learning for Monkeypox Detection with Explainable AI*. Advisor: Prof. Anichur Rahman.

RESEARCH EXPERIENCE

Lead Research Associate, ResearchUstad

Sep 2024–Present

- Develop privacy-preserving federated learning pipelines for decentralized medical and scientific datasets, with an emphasis on robust optimization across heterogeneous clients.
- Co-lead work on foundation-model adaptation, explainable AI evaluation, and LLM jailbreak robustness for trustworthy AI systems.
- Design latent-space fusion and differential privacy mechanisms to mitigate non-IID data challenges in distributed learning.
- Contribute to multiple Q1 manuscripts spanning healthcare AI, federated optimization, and security-aware generative AI.

Research Assistant, NITER (University of Dhaka)

Mar 2023–May 2025

- Designed hybrid deep learning architectures (CNNs, Transformers, LSTMs) for speech emotion recognition, flood forecasting, and medical image classification.
- Integrated federated learning, GAN-based augmentation, and SHAP explanations to improve privacy and interpretability in diagnostic models.
- Curated multi-region datasets, ran ablation studies, and performed rigorous experimental validation leading to multiple Q1 publications.

PUBLICATIONS

8 total Q1 journal articles; selected titles below are hyperlinked for quick access. Full publication list available via [Google Scholar](#).

PUBLISHED JOURNAL ARTICLES (8 TOTAL; 6 SELECTED BELOW)

* denotes first-author paper.

1. **S. H. Mahir, et al.***. Monkeypox and Skin Lesion Detection with Federated Learning, GANs and Explainable AI. *Biomedical Signal Processing and Control*. 2025.

2. **S. H. Mahir, et al.** *. Federated Flood Prediction with Multi-Region Custom Dataset, XAI and Hyper Models. *IEEE Open Journal of the Computer Society*. 2025.
3. **Jaheen Abrar, S. H. Mahir, et al.**. Bangla Speech Emotion Recognition with FL and Hybrid Models. *International Journal of Speech Technology*. 2025.
4. **A. Rahman, S. H. Mahir, et al.**. Collaborative Advancement of Cloud-Edge-Fog Computing in Smart Healthcare: Concepts, Applications, and Future Opportunities. *Cluster Computing*. 2026.
5. **M. Tashrif, S. H. Mahir, et al.**. Predicting Preterm Birth with Privacy-Preserving AI Models: Federated Learning and Explainable AI. *Egyptian Informatics Journal*. 2026.
6. **M. Kawser, S. H. Mahir, et al.**. An Explainable Framework for Dementia Risk Prediction Based on Lifestyle Data Using Machine Learning and Deep Learning. *Discover Artificial Intelligence*. 2026.

MANUSCRIPTS UNDER REVIEW / SUBMISSION (4)

* denotes first-author paper.

1. **S. H. Mahir, et al.** *. Differentiable Latent Fusion via Divergence-Aware Scaling in Federated Foundation Models. *Manuscript in submission*. 2026.
2. **S. H. Mahir, et al.** *. AgriSSFL: Federated Semi-Supervised Learning for Crop Disease Diagnosis, addressing class imbalance, non-IID data, and label scarcity jointly. *Under submission*. 2026.
3. **S. H. Mahir, et al.**. Physiological-State-Aware Probabilistic Information Fusion for Sparse-Calibration Personalization in Wearable Glucose Trend Estimation. *Information Fusion*. 2026.
4. **A. Rahman, S. H. Mahir, et al.**. Unmasking Vulnerabilities in LLMs: A Deep Dive into Jailbreak Attacks and Countermeasures. *IEEE Transactions on Neural Networks and Learning Systems (second-round review)*. 2026.

AWARDS & GRANTS

- International Research Grant Recipient (Group), Multimedia University (Malaysia) and King Saud University (Saudi Arabia), 2025.
- Best Research Paper (Runner-up), International Conference on Electrical, Computer and Communication Engineering (ICEE-ICT), 2024.
- Research Grant Awardee, Research Ustad ML Innovation Fund, 2023.

TEACHING & MENTORSHIP

Teaching Assistant, University of Dhaka

2 semesters

- Supported Algorithms I, Algorithms II, and Data Science through tutorials, grading, and one-on-one debugging help.
- Mentored 50+ students and strengthened their implementation skills in data structures and algorithmic problem solving.

ICT Instructor, Government-Funded Youth Empowerment Programme

4 months

- Taught C++ and Java fundamentals to beginner programmers through structured lectures and hands-on labs.
- Delivered 120+ hours of instruction and helped students build their first desktop applications.

TECHNICAL SKILLS

Programming: Python, C/C++, Bash

ML/DL: PyTorch, TensorFlow, scikit-learn, Pandas, NumPy

Research: Federated Learning, Foundation Models, Explainable AI, LLM Security, GANs, Transformers, LSTMs, Differential Privacy, SHAP

Tools: Git, Linux, SQL, LaTeX, Google Colab, Kaggle, Conda

SELECTED PRESENTATIONS

- **Hydro-Informatic Modeling for Flood Prediction Through Explainable AI to Interpret Water Dynamics in Bangladesh Perspective.** Presented at ICEEICT 2023.
- **Privacy-Preserving Analytics and Explainable AI in Modern Healthcare Pipelines.** Invited technical talk, NITER Computer Society Seminar Series, 2025.
- **DistB-VNET: Distributed Cluster-Based Blockchain Vehicular Ad-Hoc Networks Through SDN-NFV for Smart City Applications.** Presented at ICCIT 2024.